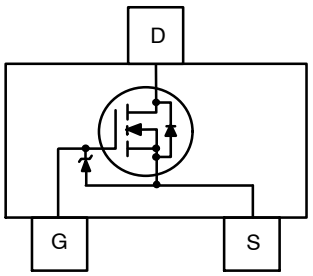
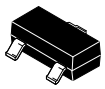
Digital FET, N-Channel

FDV301N, FDV301N-F169





SOT−23
CASE 318−08

General Description

This N−Channel logic level enhancement mode field effect transistor is produced using **onsemi's** proprietary, high cell density, DMOS technology. This very high density process is especially tailored to minimize on−state resistance. This device has been designed especially for low voltage applications as a replacement for digital transistors. Since bias resistors are not required, this one N−channel FET can replace several different digital transistors, with different bias resistor values.

Features

- 25 V, 0.22 A Continuous, 0.5 A Peak
  - $R_{DS(on)}$ = 5 Ω @ V_{GS} = 2.7 V
  - $R_{DS(on)}$ = 4 Ω @ V_{GS} = 4.5 V
- Very Low Level Gate Drive Requirements Allowing Direct Operation in 3 V Circuits. $V_{GS(th)}$ < 1.06 V
- Replace Multiple NPN Digital Transistors with One DMOS FET
- This Device is Pb−Free and Halide Free

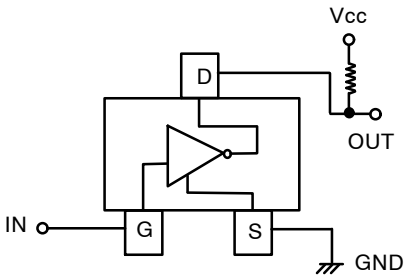
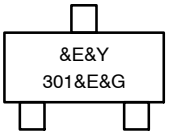


Figure 1. Inverter Application

MARKING DIAGRAM



&E	= Designates Space
&Y	= Binary Calendar Year Coding Scheme
301	= Specific Device Code
&G	= Date Code

ORDERING INFORMATION

Device	Package	Shipping†
FDV301N, FDV301N−F169	SOT−23−3 (Pb−Free, Halide−Free)	3000 / Tape & Reel

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specification Brochure, BRD8011/D.

© Semiconductor Components Industries, LLC, 2009

ABSOLUTE MAXIMUM RATINGS T_A = 25°C unless otherwise noted.

Symbol	Parameter	FDV301N	Unit
V_{DSS}, V_{CC}	Drain−Source Voltage, Power Supply Voltage	25	V
V_{GSS}, V_I	Gate−Source Voltage, V_{IN}	−0.3 to 8	V
I_D, I_O	Drain/Output Current − Continuous	0.22	A
		0.5	
P_D	Maximum Power Dissipation	0.35	W
T_J, T_{STG}	Operating and Storage Temperature Range	−55 to 150	°C

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

THERMAL CHARACTERISTICS T_A = 25°C unless otherwise noted.

Symbol	Parameter	Value	Unit
$R_{\theta JA}$	Thermal Resistance, Junction−to−Ambient	357	°C/W

INVERTER ELECTRICAL CHARACTERISTICS T_A = 25°C unless otherwise noted.

Symbol	Parameter	Test Conditions	Min	Typ	Max	Unit
$I_{O(off)}$	Zero Input Voltage Output Current	V_{CC} = 20 V, V_I = 0 V	−	−	1	μA
$V_{I(off)}$	Input Voltage	V_{CC} = 5 V, I_O = 10 μA	−	−	0.5	V
$V_{I(on)}$		V_O = 0.3 V, I_O = 0.005 A	1	−	−	
$R_{O(on)}$	Output to Ground Resistance	V_I = 2.7 V, I_O = 0.2 A	−	4	5	Ω

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

ELECTRICAL CHARACTERISTICS T_A = 25°C unless otherwise noted.

Symbol	Parameter	Test Conditions	Min	Typ	Max	Unit
OFF CHARACTERISTICS						
BV_{DSS}	Drain−Source Breakdown Voltage	V_{GS} = 0 V, I_D = 250 μA	25	−	−	V
$\Delta BV_{DSS}/\Delta T_J$	Breakdown Voltage Temp. Coefficient	I_D = 250 μA, Referenced to 25°C	−	25	−	mV/°C
I_{DSS}	Zero Gate Voltage Drain Current	V_{DS} = 20 V, V_{GS} = 0 V	−	−	1	μA
		V_{DS} = 20 V, V_{GS} = 0 V, T_J = 55°C	−	−	10	
I_{GSS}	Gate − Body Leakage Current	V_{GS} = 8 V, V_{DS} = 0 V	−	−	100	nA
ON CHARACTERISTICS						
$\Delta V_{GS(th)}/\Delta T_J$	Gate Threshold Voltage Temp. Coefficient	I_D = 250 μA, Referenced to 25°C	−	−2.1	−	mV/°C
$V_{GS(th)}$	Gate Threshold Voltage	V_{DS} = V_{GS}, I_D = 250 μA	0.70	0.85	1.06	V
$R_{DS(on)}$	Static Drain−Source On−Resistance	V_{GS} = 2.7 V, I_D = 0.2 A	−	3.8	5	Ω
		V_{GS} = 2.7 V, I_D = 0.2 A, T_J = 125°C	−	6.3	9	
		V_{GS} = 4.5 V, I_D = 0.4 A	−	3.1	4	
$I_{D(on)}$	On−State Drain Current	V_{GS} = 2.7 V, V_{DS} = 5 V	0.2	−	−	A
g_{FS}	Forward Transconductance	V_{DS} = 5 V, I_D = 0.4 A	−	0.2	−	S

Share Feedback
Your Opinion Matters

ELECTRICAL CHARACTERISTICS T_A = 25°C unless otherwise noted. (continued)

Symbol	Parameter	Test Conditions	Min	Typ	Max	Unit
DYNAMIC CHARACTERISTICS						
C_{iss}	Input Capacitance	V_{DS} = 10 V, V_{GS} = 0 V, f = 1.0 MHz	–	9.5	–	pF
C_{oss}	Output Capacitance		–	6	–	
C_{rss}	Reverse Transfer Capacitance		–	1.3	–	
SWITCHING CHARACTERISTICS (Note 1)						
$t_{D(on)}$	Turn − On Delay Time	V_{DD} = 6 V, I_D = 0.5 A, V_{GS} = 4.5 V, R_{GEN} = 50 Ω	–	3.2	8	ns
t_r	Turn − On Rise Time		–	6	15	
$t_{D(off)}$	Turn − Off Delay Time		–	3.5	8	
t_f	Turn − Off Fall Time		–	3.5	8	
Q_g	Total Gate Charge	V_{DS} = 5 V, I_D = 0.2 A, V_{GS} = 4.5 V	–	0.49	0.7	nC
Q_{gs}	Gate−Source Charge		–	0.22	–	
Q_{gd}	Gate−Drain Charge		–	0.07	–	
DRAIN−SOURCE DIODE CHARACTERISTICS AND MAXIMUM RATINGS						
I_S	Maximum Continuous Drain−Source Diode Forward Current		–	–	0.29	A
V_{SD}	Drain−Source Diode Forward Voltage	V_{GS} = 0 V, I_S = 0.29 A (Note 1)	–	0.8	1.2	V

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

1. Pulse Test: Pulse Width ≤ 300 μs, Duty Cycle ≤ 2.0%.

TYPICAL CHARACTERISTICS

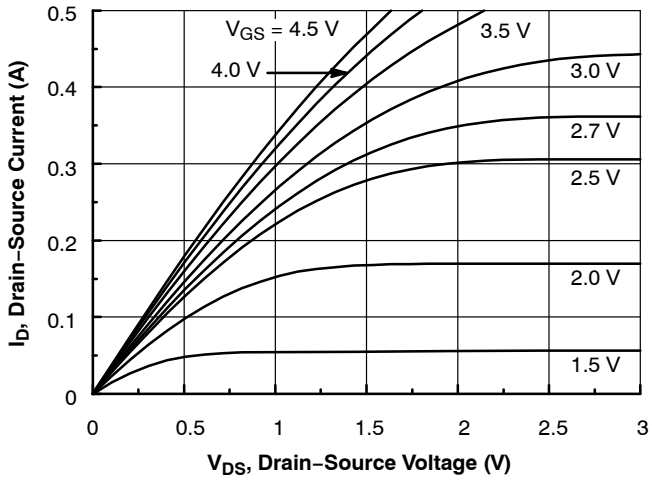


Figure 2. On−Region Characteristics

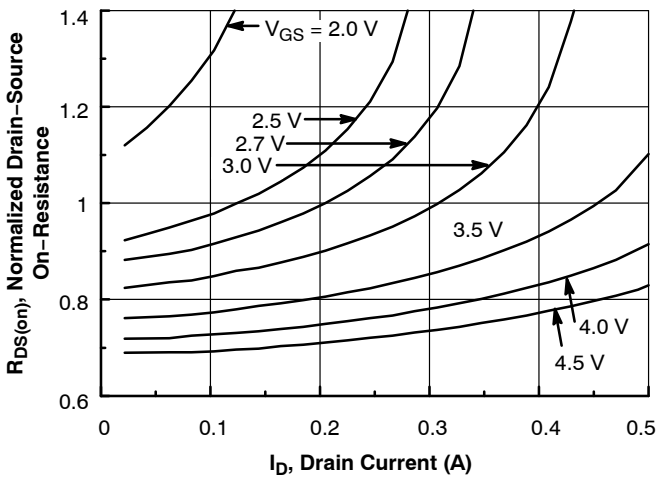


Figure 3. On−Resistance Variation with Drain Current and Gate Voltage

Share Feedback
Your Opinion Matters

TYPICAL PERFORMANCE CHARACTERISTICS (continued)

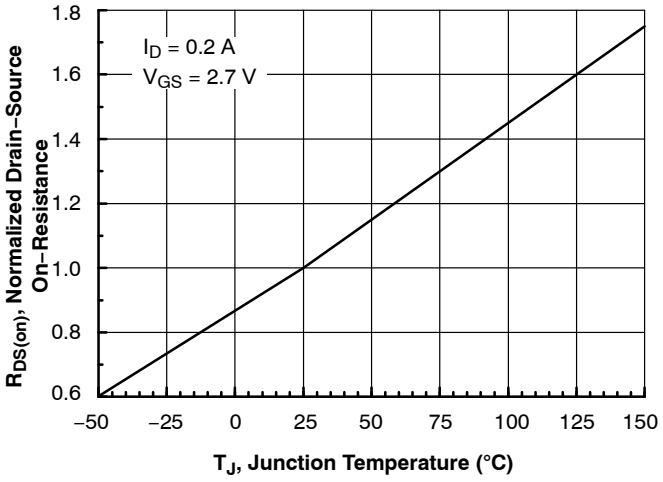


Figure 4. On−Resistance Variation with Temperature

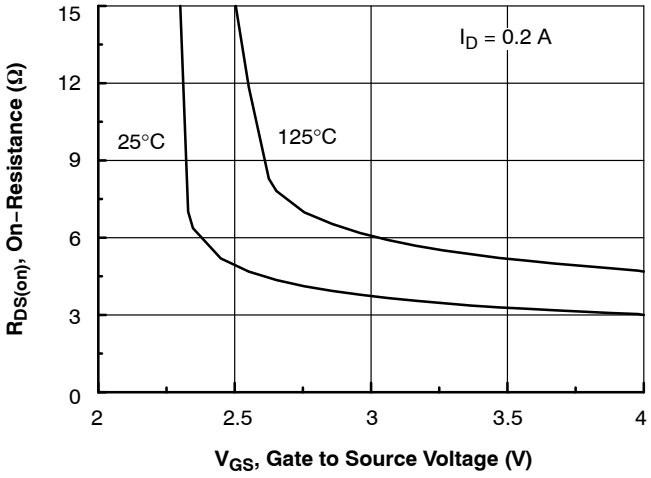


Figure 5. On Resistance Variation with Gate−To−Source Voltage

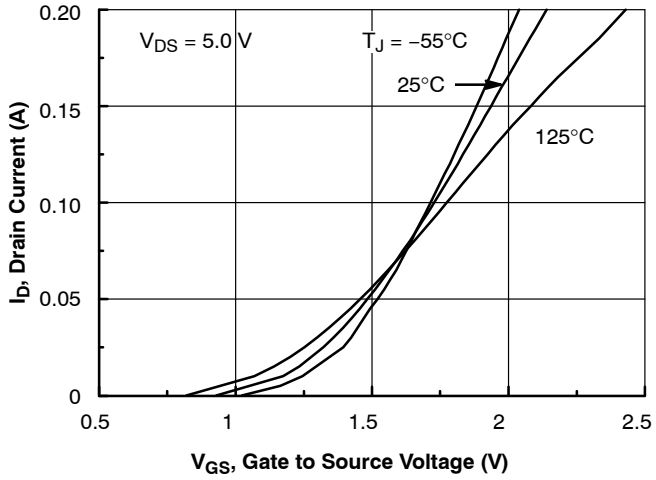


Figure 6. Transfer Characteristics

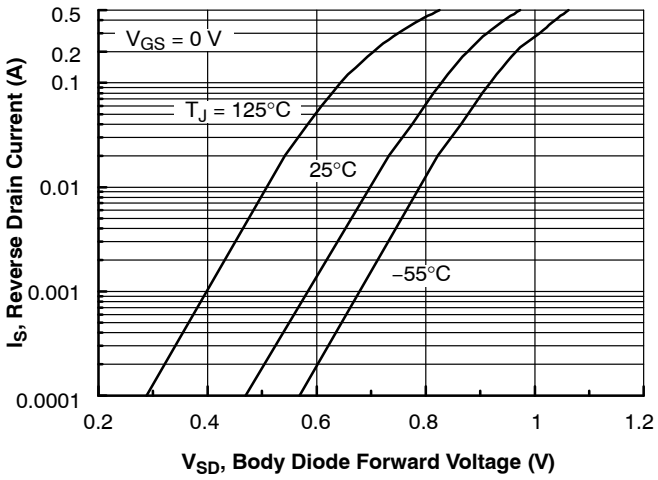


Figure 7. Body Diode Forward Voltage Variation with Source Current and Temperature

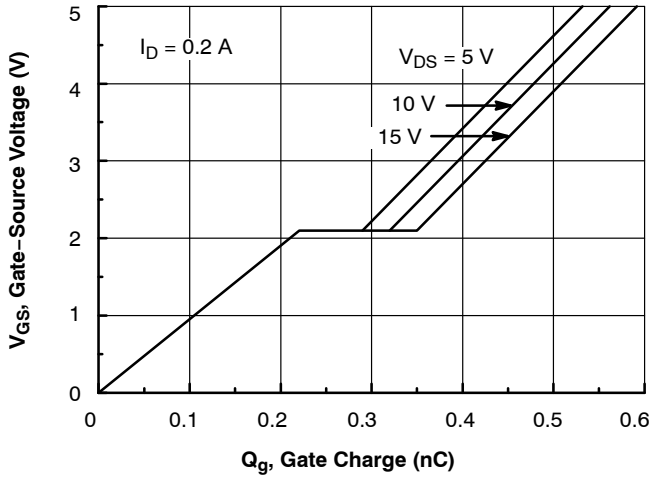


Figure 8. Gate Charge Characteristics

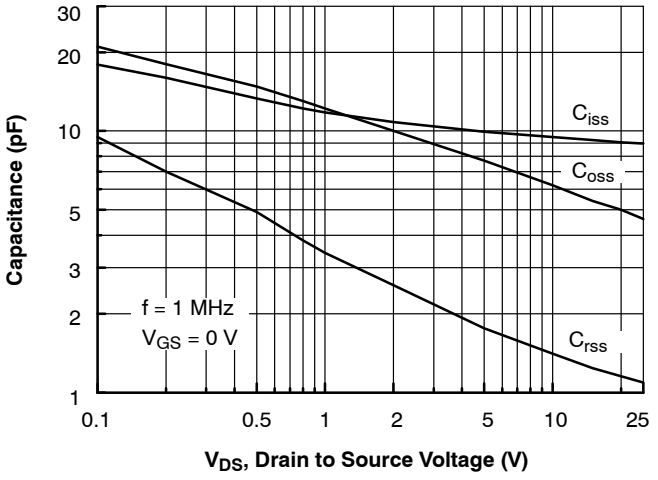


Figure 9. Capacitance Characteristics

Share Feedback
Your Opinion Matters

TYPICAL PERFORMANCE CHARACTERISTICS (continued)

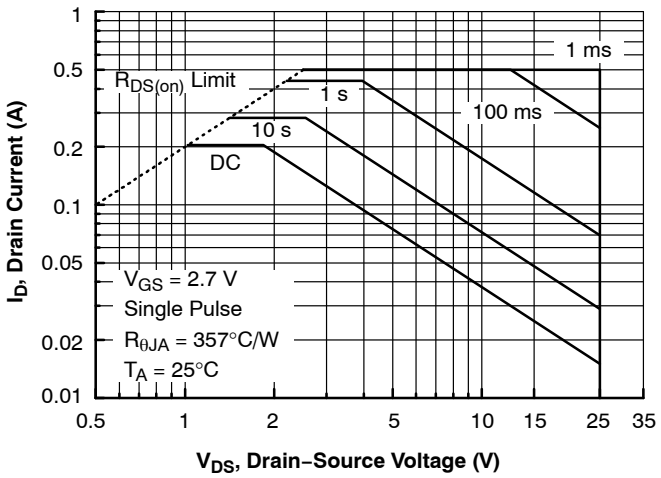


Figure 10. Maximum Safe Operating Area

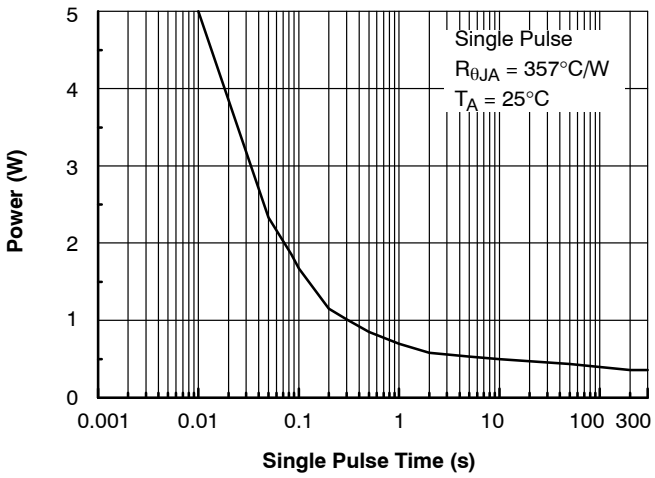


Figure 11. Single Pulse Maximum Power Dissipation

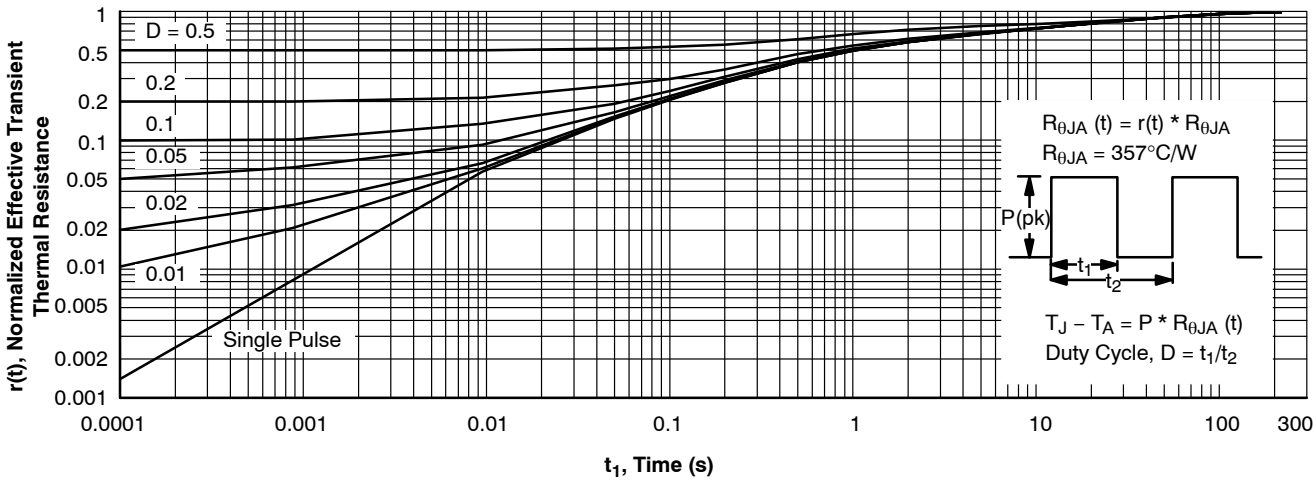
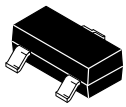


Figure 12. Transient Thermal Response Curve

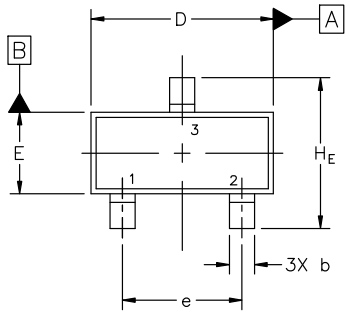
Share Feedback
Your Opinion Matters



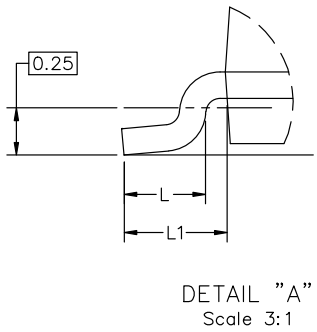
SCALE 4:1

SOT−23 (TO−236) 2.90x1.30x1.00 1.90P
CASE 318
ISSUE AU

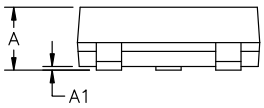
DATE 14 AUG 2024



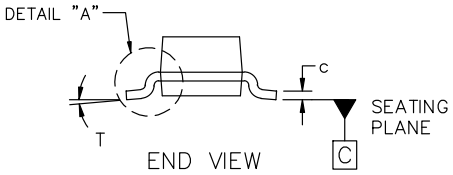
TOP VIEW



DETAIL "A"
Scale 3:1



SIDE VIEW

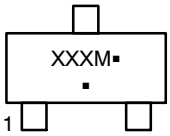


END VIEW

MILLIMETERS			
DIM	MIN	NOM	MAX
A	0.89	1.00	1.11
A1	0.01	0.06	0.10
b	0.37	0.44	0.50
c	0.08	0.14	0.20
D	2.80	2.90	3.04
E	1.20	1.30	1.40
e	1.78	1.90	2.04
L	0.30	0.43	0.55
L1	0.35	0.54	0.69
H_E	2.10	2.40	2.64
T	0°	−−−	10°

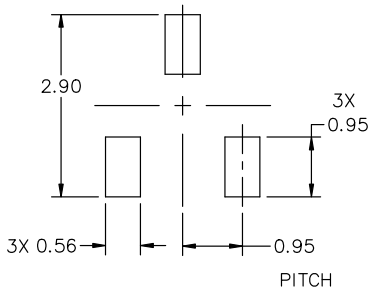
NOTES:
1. DIMENSIONING AND TOLERANCING PER ASME Y14.5M, 2018.
2. CONTROLLING DIMENSIONS: MILLIMETERS.
3. MAXIMUM LEAD THICKNESS INCLUDES LEAD FINISH. MINIMUM LEAD THICKNESS IS THE MINIMUM THICKNESS OF THE BASE MATERIAL.
4. DIMENSIONS D AND E DO NOT INCLUDE MOLD FLASH, PROTRUSIONS, OR GATE BURRS.

GENERIC
MARKING DIAGRAM*



XXX = Specific Device Code
M = Date Code
▪ = Pb−Free Package

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, "G" or microdot "▪", may or may not be present. Some products may not follow the Generic Marking.



RECOMMENDED
MOUNTING FOOTPRINT

* For additional information on our Pb−Free strategy and soldering details, please download the onsemi Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

STYLES ON PAGE 2

onsemi and onsemi are trademarks of Semiconductor Components Industries, LLC dba **onsemi** or its subsidiaries in the United States and/or other countries. **onsemi** reserves the right to make changes without further notice to any products herein. **onsemi** makes no warranty, representation or guarantee regarding the suitability of its products for any particular purpose, nor does **onsemi** assume any liability arising out of the application or use of any product or circuit, and specifically disclaims any and all liability, including without limitation special, consequential or incidental damages. **onsemi** does not convey any license under its patent rights nor the rights of others.

© Semiconductor Components Industries, LLC, 2019

SOT−23 (TO−236) 2.90x1.30x1.00 1.90P
CASE 318
ISSUE AU

DATE 14 AUG 2024

STYLE 1 THRU 5:
CANCELLED

STYLE 6:
PIN 1. BASE
2. EMITTER
3. COLLECTOR

STYLE 7:
PIN 1. EMITTER
2. BASE
3. COLLECTOR

STYLE 8:
PIN 1. ANODE
2. NO CONNECTION
3. CATHODE

STYLE 9:
PIN 1. ANODE
2. ANODE
3. CATHODE

STYLE 10:
PIN 1. DRAIN
2. SOURCE
3. GATE

STYLE 11:
PIN 1. ANODE
2. CATHODE
3. CATHODE−ANODE

STYLE 12:
PIN 1. CATHODE
2. CATHODE
3. ANODE

STYLE 13:
PIN 1. SOURCE
2. DRAIN
3. GATE

STYLE 14:
PIN 1. CATHODE
2. GATE
3. ANODE

STYLE 15:
PIN 1. GATE
2. CATHODE
3. ANODE

STYLE 16:
PIN 1. ANODE
2. CATHODE
3. CATHODE

STYLE 17:
PIN 1. NO CONNECTION
2. ANODE
3. CATHODE

STYLE 18:
PIN 1. NO CONNECTION
2. CATHODE
3. ANODE

STYLE 19:
PIN 1. CATHODE
2. ANODE
3. CATHODE−ANODE

STYLE 20:
PIN 1. CATHODE
2. ANODE
3. GATE

STYLE 21:
PIN 1. GATE
2. SOURCE
3. DRAIN

STYLE 22:
PIN 1. RETURN
2. OUTPUT
3. INPUT

STYLE 23:
PIN 1. ANODE
2. ANODE
3. CATHODE

STYLE 24:
PIN 1. GATE
2. DRAIN
3. SOURCE

STYLE 25:
PIN 1. ANODE
2. CATHODE
3. GATE

STYLE 26:
PIN 1. CATHODE
2. ANODE
3. NO CONNECTION

STYLE 27:
PIN 1. CATHODE
2. CATHODE
3. CATHODE

STYLE 28:
PIN 1. ANODE
2. ANODE
3. ANODE

onsemi and ONSEMI are trademarks of Semiconductor Components Industries, LLC dba **onsemi** or its subsidiaries in the United States and/or other countries. **onsemi** reserves the right to make changes without further notice to any products herein. **onsemi** makes no warranty, representation or guarantee regarding the suitability of its products for any particular purpose, nor does **onsemi** assume any liability arising out of the application or use of any product or circuit, and specifically disclaims any and all liability, including without limitation special, consequential or incidental damages. **onsemi** does not convey any license under its patent rights nor the rights of others.

© Semiconductor Components Industries, LLC, 2019

onsemi, ONSEMI , and other names, marks, and brands are registered and/or common law trademarks of Semiconductor Components Industries, LLC dba "**onsemi**" or its affiliates and/or subsidiaries in the United States and/or other countries. **onsemi** owns the rights to a number of patents, trademarks, copyrights, trade secrets, and other intellectual property. A listing of **onsemi**'s product/patent coverage may be accessed at www.onsemi.com/site/pdf/Patent−Marking.pdf. **onsemi** reserves the right to make changes at any time to any products or information herein, without notice. The information herein is provided "as−is" and **onsemi** makes no warranty, representation or guarantee regarding the accuracy of the information, product features, availability, functionality, or suitability of its products for any particular purpose, nor does **onsemi** assume any liability arising out of the application or use of any product or circuit, and specifically disclaims any and all liability, including without limitation special, consequential or incidental damages. Buyer is responsible for its products and applications using **onsemi** products, including compliance with all laws, regulations and safety requirements or standards, regardless of any support or applications information provided by **onsemi**. "Typical" parameters which may be provided in **onsemi** data sheets and/or specifications can and do vary in different applications and actual performance may vary over time. All operating parameters, including "Typicals" must be validated for each customer application by customer's technical experts. **onsemi** does not convey any license under any of its intellectual property rights nor the rights of others. **onsemi** products are not designed, intended, or authorized for use as a critical component in life support systems or any FDA Class 3 medical devices or medical devices with a same or similar classification in a foreign jurisdiction or any devices intended for implantation in the human body. Should Buyer purchase or use **onsemi** products for any such unintended or unauthorized application, Buyer shall indemnify and hold **onsemi** and its officers, employees, subsidiaries, affiliates, and distributors harmless against all claims, costs, damages, and expenses, and reasonable attorney fees arising out of, directly or indirectly, any claim of personal injury or death associated with such unintended or unauthorized use, even if such claim alleges that **onsemi** was negligent regarding the design or manufacture of the part. **onsemi** is an Equal Opportunity/Affirmative Action Employer. This literature is subject to all applicable copyright laws and is not for resale in any manner.

ADDITIONAL INFORMATION

TECHNICAL PUBLICATIONS:
Technical Library: www.onsemi.com/design/resources/technical−documentation
onsemi Website: www.onsemi.com

ONLINE SUPPORT: www.onsemi.com/support
For additional information, please contact your local Sales Representative at
www.onsemi.com/support/sales